For Examiner's Use

Section A

Answer **all** questions in the spaces provided on the question paper.

1 When copper is added to an acid, what happens depends on the acid used and the conditions. For example, when copper reacts with dilute nitric acid, copper nitrate (V) and NO is produced. However if copper reacts with concentrated nitric acid, NO_2 is produced instead in addition to copper nitrate (V).

(a) Construct the half-equation for the reduction of nitrate to NO and hence write an balanced equation for this reaction.

$NO_3^- + 4H^+ + 3e \rightarrow NO + 2H_2O$

$3Cu + 8HNO_3 \rightarrow 2\ NO + 3Cu(NO_3)_2 + 4H_2O$

Or

$3Cu + 8H^+ + 2NO_3^- \rightarrow 2NO + 3Cu^{2+} + 4H_2O$

[2]

(b) If 10g of copper was reacted with dilute nitric acid, calculate:

(i) Volume of NO produced, measured at room temperature and pressure.

nCu = 10/63.5 = 0.157
nNO = 2/3 x (0.157) = 0.105
Vol of NO = 0.105 x 24 = 2.52 dm^3

[2]

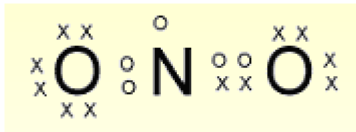
(ii) Volume of 1.5 mol dm^{-3} nitric acid needed for complete reaction.

$nHNO_3$ = 8/3 x (0.157) = 0.419
Vol of HNO_3 = 0.419 / 1.5 = 0.279 dm^3

[2]

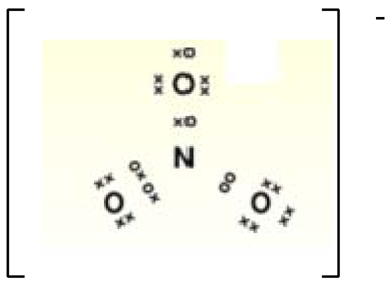
1 (c) Draw the dot and cross diagram for the following species and predict their shapes.

(i) NO_2



bent [2]

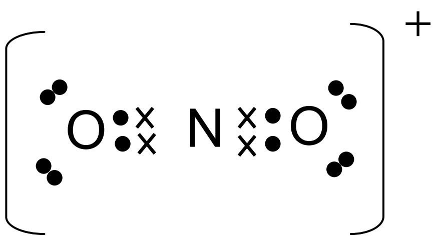
(ii) NO_2^-



bent

[2]

(iii) NO_2^+



linear

[2]

(d) How will the bond angles in **(c)(i)** and **(c)(ii)** compare? Explain.

- bond angle for NO_2^- will be smaller than that of NO_2
- a lone pair orbital in NO_2^- exerts a bigger repulsion than an orbital with one electron in NO_2

[2]

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1 **(e)** At room temperature and pressure NO_2 dimerises to form dinitrogen tetraoxide, N_2O_4, as shown below:

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -ve$$

(i) NO behaves similarly. What will it form and why does this happen?

- It will dimerise to form N_2O_2
- presence on odd unpaired electron can be used in covalent bond formation

[2]

(ii) Write the expression for the equilibrium constant, K_C for the above equilibrium.

- $Kc = [N_2O_4] / [NO_2]^2$

[1]

(iii) At 298 K and 101 kPa, 1.00g of the mixture of these two gases at equilibrium occupying a volume of 0.317 dm^3 is found to show an average Mr of 77.3. The average Mr of the mixture can be computed using the following expression,

$$\text{Ave Mr} = \frac{[n_{equilm}(NO_2) \times Mr(NO_2)] + [n_{equilm}(N_2O_4) \times Mr(N_2O_4)]}{\text{Total number of moles at equilibrium}}$$

Using the expression, calculate the concentration of each gas in the mixture at equilibrium and hence determine the K_C.

Fill in the table below and using the expression given above, solve for the value of *y*.

	NO_2	N_2O_4
Initial/ mol	1/46	0
Change/ mol	$-2y$	$+y$
Equilibrium/ mol	1/46 – 2y	y

$$\frac{[(1/46 - 2y) \times 46] + [y \times 92]}{(1/46 - y)} = 77.3$$

y = 0.00878

[1]

1 (e) (iv) Hence, determine the value of Kc.

n_{equilm} (NO_2) = y = 0.00878 moles
n_{equilm} (N_2O_4) = 0.00413 moles

$[N_2O_4]$ = 0.00878 / 0.317 = 0.0277 mol dm^{-3}
$[NO_2]$ = 0.00413 / 0.317 = 0.0130 mol dm^{-3}

Kc = 0.0277 / $(0.0130)^2$ = 164 mol^{-1} dm^3

(v) Deduce qualitatively the effect on the average Mr of this gaseous mixture when

the pressure is increased

- Ave Mr will increase
- Equilm shifts to the right (in the direction of fewer particles)

[2]

the temperature Is increased

- Ave Mr will decrease
- Equilm shifts to the left (in the direction of absorption of heat)

[2]

[Total: 25]

2 Compound **W** reacts in two steps to form compound **Y**.

$$\underset{\textbf{W}}{C_4H_9Br} \xrightarrow{\textbf{Step 1}} \underset{\textbf{X}}{C_4H_{10}O} \xrightarrow{\textbf{Step 2}} \underset{\textbf{Y}}{C_4H_8}$$

(a) What are the reagents and conditions needed for steps **1** and **2**?

- Step 1: NaOH(aq) and heat
- Step 2: Concentrated H_2SO_4 and 170^oC

[2]

(b) (i) Compound **Y** is formed as a single isomer in the above reaction. If **Y** can be oxidised to form a carboxylic acid, suggest a structural formula for **W** and the corresponding **X**.

- W is $CH_3CH_2CH_2CH_2Br$
- X is $CH_3CH_2CH_2CH_2OH$

[2]

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2 **(b)** **(ii)** State the reagent and condition needed to bring about the oxidation of **Y** and write a balanced equation for the oxidation of **Y**. [2]

- Acidified managanate and heat
- $CH_3CH_2CH=CH_2 + 5[O] \rightarrow CH_3CH_2COOH + CO_2 + H_2O$

(c) **(i)** Another isomer of **W** also yields a single isomer of the product Y in the above sequence. What are the possible structures for **W**?

- $(CH_3)_3CBr$
- $(CH_3)_2CHCH_2Br$

[2]

(ii) If the product **Y** from **(c)(i)** is now oxidised, what will be the resulting organic product?

- CH_3COCH_3

[1]

(b) **(iii)** **W** can be converted into **Y** in a single step. What reagent and conditions are needed for this single step conversion?

- Concentrated KOH in alcoholic medium and heat

[1]

[Total: 10]

3 Iodine reacts with propanone as follows,

$$I_2 + CH_3COCH_3 \rightarrow CH_3COCH_2I + HI$$

A possible mechanism for this reaction is

$CH_3\text{-}C\text{=}O + H^+ \rightarrow CH_3\text{-}C\text{=}OH^+$ (each with CH_3 on C) Slow

$CH_3\text{-}C\text{=}OH^+$ (with CH_3 on C) $\rightarrow CH_3\text{-}C\text{-}OH + H^+$ (with $=CH_2$ on C) fast

$CH_3\text{-}C\text{-}OH$ (with $=CH_2$ on C) $+ I_2 \rightarrow CH_3\text{-}C\text{=}O + HI$ (with CH_2I on C) fast

(a) Write a rate equation for this reaction based on the above mechanism.

- Rate = k $[CH_3COCH_3]$ $[H^+]$

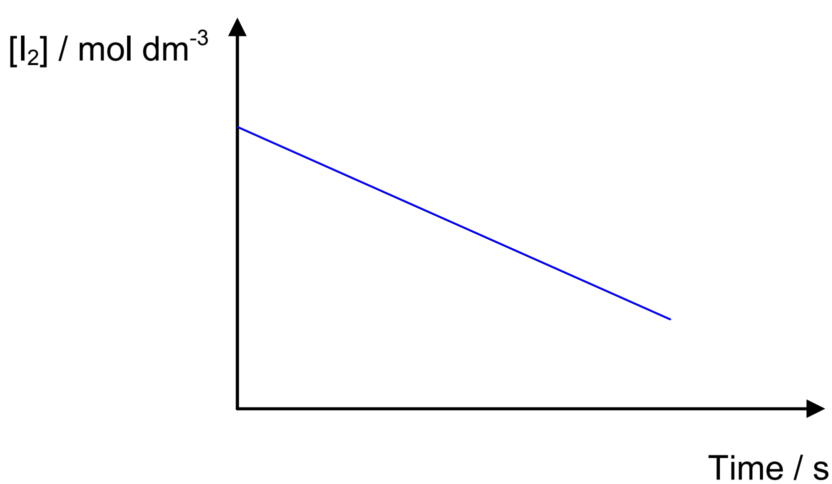
[1]

(b) What evidence indicates that the reaction is acid catalysed?

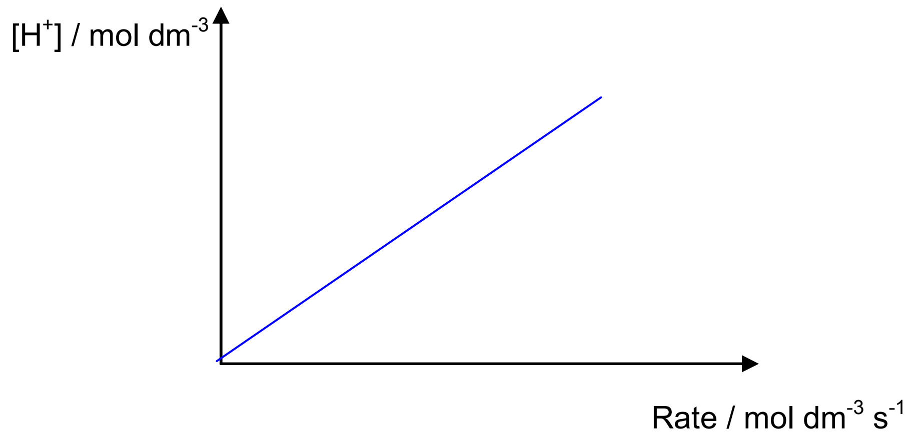
- H^+ is present at the start and end of the reaction

[1]

(c) On the axes below sketch the graphs you will obtain for the above reaction.



3 (c)



[2]

(d) Iodine and propanone reacts differently if NaOH(aq) is present instead of H$^+$(aq).
How will the reaction be different?

- CHI_3 and $CH_3COO^-Na^+$ will form

[1]

[Total: 5]

4 Sulfur is abundant in the Earth's crust. It occurs as elemental sulfur, as mineral sulfides and sulfates, and as organic sulfur compounds in oil and coal. However, sulfur is also a significant source of pollutant when it burns.

(a) Describe the observation expected when

(i) the element reacts with oxygen

(ii) the resulting gaseous product bubbles into water containing universal indicator.

Write equations for the reactions described. [5]

Sulfur burns with a blue flame which gives SO_2 and is oxidised to SO_3 in excess oxygen.

$S(s) + O_2(g) \rightarrow SO_2(g)$
$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

SO_2 and SO_3 dissolve readily in water to give an acidic solution which turns universal indicator red.

$SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$
$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$

(b) Sulfur dioxide is also a good reductant. In the presence of water, sulfur dioxide is able to decolorize substances.

A solution of NH_4VO_3 in dilute H_2SO_4 can be reduced by SO_2 to give a deep blue solution. A 25.0 cm^3 of a solution containing 5.85 g of NH_4VO_3 per dm^3 were reduced and then re-oxidised to the orignal state by 12.5 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$. Deduce the oxidation state of the Vanadium in the deep blue solution. [5]

No of moles of VO_3^- $= \frac{25.0}{1000} \times 5.85 \div (14.0 + 4.0 + 50.9 + 3 \times 16.0)$

= 0.00125 mol

$MnO_4^- \equiv 5e^-$

No of moles of electrons removed by MnO_4^- $= \frac{12.5}{1000} \times 0.0200 \times 5$

= 0.00125 mol

Therefore, the change in oxidation state = $\frac{0.00125}{0.00125}$ **= 1**

and thus the oxidation state of Vanadium in the deep blue solution is +4.

(c) Thiosulfate occurs naturally and is produced by certain biochemical processes, but it can also be made from the reaction below.

(i) The overall equation for the formation of thiosulfate ion from sulfite ion and sulfur in an alkaline solution is given below.
$SO_3^{2-}(aq) + S(s) \rightarrow S_2O_3^{2-}(aq)$

Write equations for the half-reactions that occur. [2]

Oxidation: $2S(s) + 6OH^- \rightarrow \rightarrow S_2O_3^{2-} + 3H_2O + 4e$
Reduction: $2SO_3^{2-}(aq) + 3H_2O + 4e \rightarrow S_2O_3^{2-} + 6OH^-$

(ii) In an acidic medium, the thiosulfate is oxidised by $Fe^{3+}(aq)$, which itself is reduced to $Fe^{2+}(aq)$ according to the half-equations below.

$S_2O_3^{2-} + 5H_2O \rightarrow 2SO_4^{2-} + 10H^+ + 8e^-$
$Fe^{3+} + e^- \rightarrow Fe^{2+}$

In an experiment, 25.0 cm^3 of 0.100 mol dm^{-3} $S_2O_3^{2-}$ requires 20.0 cm^3 of the Fe^{3+} solution for complete reaction. What is the concentration of the Fe^{3+} solution? [4]

No of moles of $S_2O_3^{2-}$ = $\frac{25}{1000} \times 0.100$ = $\underline{2.50 \times 10^{-3}}$ mol

$S_2O_3^{2--} \equiv 8Fe^{3+}$

No of moles of Fe^{3+} = 8 x $\underline{2.50 \times 10^{-3}}$ = $\underline{0.0200}$ mol

Concentration of Fe^{3+} = (0.0200 /20)x 1000
= 1.00 mol dm^{-3}

(d) Use your knowledge of the chemistry of the Periodic Table to answer the following questions. Where possible, give reasons for your answers. [4]

(i) How does the atomic radius of sulfur compare with that of oxygen?

The atomic radius of sulfur is bigger than that of oxygen.

As sulfur is below oxygen in Group VI, it has an extra (principal quantum) shell of electrons. Hence it is bigger.

(ii) How does the ionisation energy of sulfur compare with that of phosphorus?

Lower.

The electron to be removed from sulfur is paired. Due to inter-electronic repulsion, it was easier to remove.

Illustration with electronic configuration is expected.

[Total: 20]

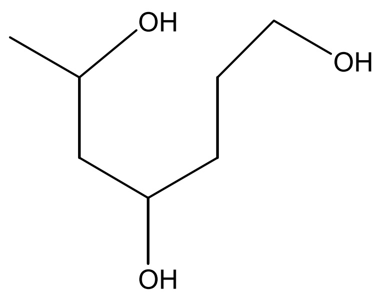
5 (a) Compound **A** ($C_{10}H_{16}$) contains a 6-membered carbon cyclic ring.

On hydrogenation, in the presence of nickel catalyst, compound **B** ($C_{10}H_{20}$) is formed. On hydration, compound **A** forms compound **C**, a diol, which cannot undergo further oxidation by hot acidified potassium dichromate(VI).

Compound **A** can be oxidised by hot acidified potassium manganate(VII) to compounds **D** and **E**.

Compound **D** contains 62.1 % C, 10.0 % H and 27.6% O. Compound **D** produces an orange precipitate on undergoing condensation with 2,4-dinitrophenylhydrazine.

Compound **E** can be reduced to give the compound **F** below.



F

[10]

Suggest, with reasons, the structures of compounds **A**, **B**, **C, D**, and **E**.

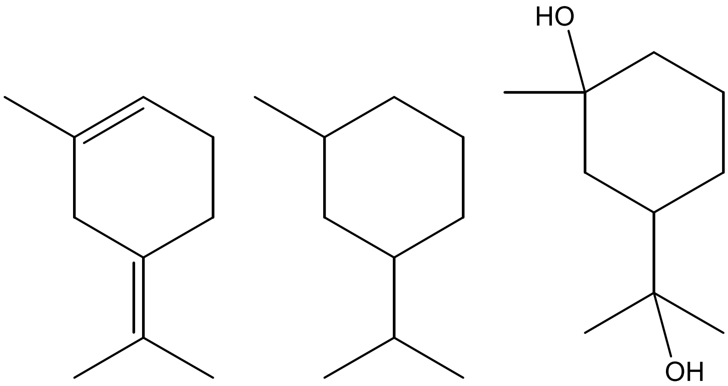
A undergoes hydrogenation to form B. The number of H atoms increases by 4. A contains 2 C=C bonds.

After hydration, C, the diol cannot undergo further oxidation. C contains 2 tertiary alcohols.

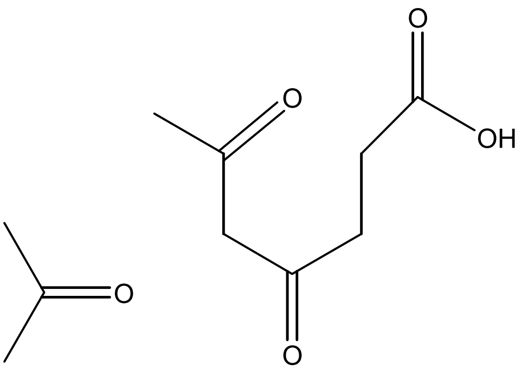
Based on calculation, the empirical formula of D is C_3H_6O.

D contains a carbonyl group as it undergoes condensation with 2,4-DNPH.

From the compound after reduction, E must contain 2 ketones and 1 carboxylic acid group as E is formed from the cleavage of C=C via vigorous oxidation.



A **B** **C**



D **E**

(b) Compound **F** can be used as a fuel as well as a solvent.

An experiment was carried out as follows to determine the standard enthalpy change of combustion of **F**, contained in a gas cylinder.

A large beaker of water was heated by burning compound **F**. The temperature rise was recorded. The cylinder was weighed before and after the experiment to determine the mass of **F** used. The following results were obtained.

Mass of **F** used/ g	2.5
Mass of water heated/ g	500
Initial temperature / ^{o}C	30
Final temperature/ ^{o}C	90

(i) Define the term enthalpy change of combustion. [1]

The standard enthalpy of combustion is the enthalpy change when 1 mole of a substance is completely burnt in oxygen at 298 K and 1 atm pressure.

(ii) Write a balanced equation for the combustion of **F** including state symbols. [1]

$C_7H_{16}O_3$ (l) + 19/2 O_2 (g) $\rightarrow$ 7 CO_2 (g) + 8 H_2O (l)

(iii) The use of the Data Booklet is relevant to this question.

Use the data given to calculate the enthalpy change of combustion of compound **F**. [3]

$\Delta T = 90 - 30 = 60\ ^{o}C$

$q = mc\Delta T = (500)(4.2)(60) = -126$ kJ

$\Delta H = 126 / (2.5/148) = -7460$ kJ mol^{-1}

(iv) Use the bond energies given in the Data Booklet to calculate another value of the enthalpy change of combustion of compound **F**. [3]

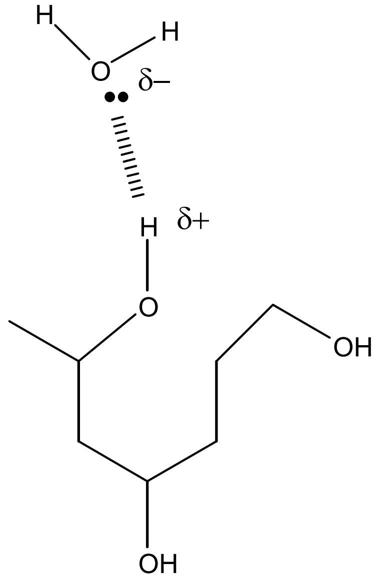
Bonds broken			**Bonds formed**		
C-H	**13**	**+410**	**C=O**	**14**	**-740**
C-C	**6**	**+350**	**O-H**	**16**	**-460**
C-O	**3**	**+360**			
O-H	**3**	**+460**			
O=O	**19/2**	**+496**			
Sum	**+14602**		**Sum**	**-17720**	

$\Delta H = (+14602) + (-17720) = -3120$ kJ mol^{-1}

(v) Suggest a reason for the discrepancy between the calculated values in (b)(iii) and (b)(iv). [1]

The value in (b)(iii) is less exothermic than the value in b(iv) because all the species in b(iv) are gaseous. Enthalpy of vaporisation is not considered in (b)(iv).

(vi) Compound **F** is miscible with water. Draw a diagram, including dipoles, to show the hydrogen bonding between a molecule of **F** and a molecule of water. [1]



[Total: 20]

6 (a) Compound **G** is a sweet-smelling cyclic compound of molecular formula $C_6H_8O_4$.

Compound **G** reacts with dilute sulfuric acid on heating to yield only one product, **H**. Compound **H** reacts with sodium carbonate to give a colourless and odourless gas which gives white precipitate with limewater and with PCl_5 to form dense white fumes.

Upon reduction, compound **H** forms compound **K**, of molecular formula $C_3H_8O_2$. Compound **K** produces a yellow precipitate on reaction with alkaline aqueous sodium hydroxide. 1 mole of compound **K** reacts with 2 moles of sodium to yield 1 mole of hydrogen gas.

Compound **G** can also be reduced by $LiAlH_4$ to **K**.

Suggest, with reasons, the structures of compounds **G**, **H** and **K**. [9]

G is an ester as it is sweet smelling.

G undergoes acidic hydrolysis to yield only one product, and hence H must contain both the carboxylic acid and the alcohol functional group.

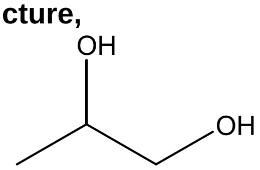
H gives CO_2 with Na_2CO_3, and hence H is carboxylic acid

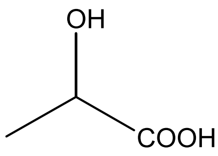
H gives HCl with PCl_5, and hence H contains the alcohol functional group or the carboxylic functional group.

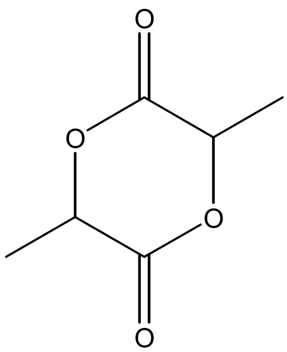
Since H undergoes reduction to form K which produces yellow ppt, K must have –C(OH)HCH$_3$ structure.

1 mole of K reacts with 2 moles of sodium to yield 1 mole of hydrogen gas, K has two alcohol function group.

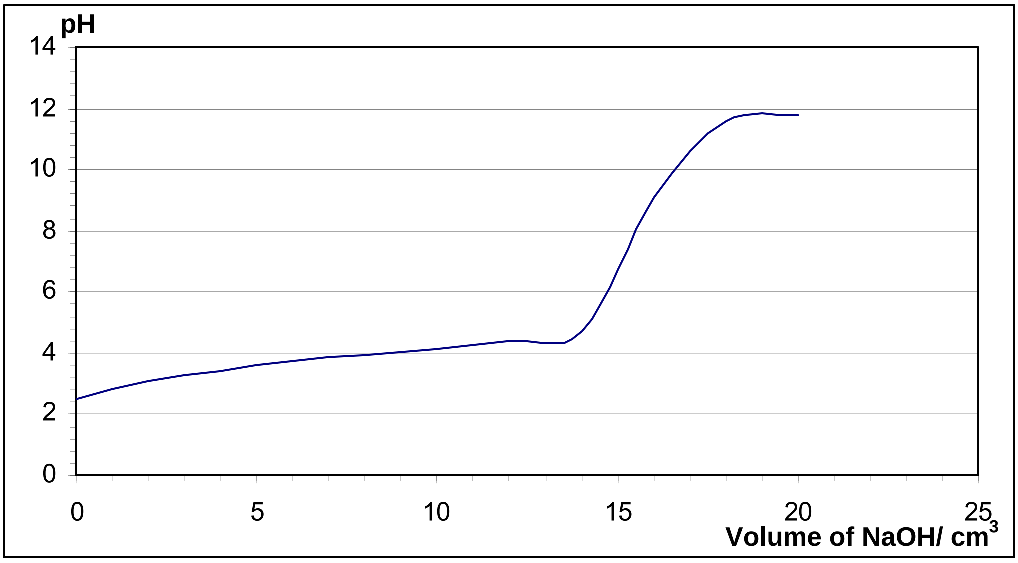
Since K has the molecular formula $C_3H_8O_2$ and –C(OH)HCH$_3$ structure,

K is 

Hence, H is 

Therefore, G is 

(b) When 10.0 cm^3 of a solution of **H** was titrated against 0.050 mol dm^{-3} of sodium hydroxide, the following graph is obtained.



(i) For this titration, suggest a suitable indicator to determine the end-point of the reaction. [2]

This is a weak acid-strong base titration, and hence the equivalence point is above pH 7. A suitable indicator would be phenolphthalein.

(ii) Calculate the concentration of **H** in the solution in mol dm^{-3}. [3]

From the graph, the equivalence volume is 15.5 cm^3.
[Accept value between 15 to 16 cm^3]

No of moles of NaOH = 15.5 x 0.050/1000 = 7.75 x 10^{-4} mol
No of moles of H = 7.75 x 10^{-4} mol
[H] = 7.75 x 10^{-4} mol / 10 x 1000 = 0.0775 mol dm^{-3}

(iii) A buffer mixture is obtained when **H** is titrated against sodium hydroxide. Explain what is meant by the term buffer solution. [2]

A buffer solution is a mixture of the weak acid and its salt. It is a solution whose pH remains approximately constant when a small amount of acid or alkali is added to it.

(iv) State the volume, with reason, at which maximum buffer capacity occurs when sodium hydroxide added. [2]

Maximum buffer capacity occurs at half of the equivalence volume = 7.75 cm^3 (reading off the graph)

Note: range from 7.50 to 8 cm^3 is acceptable.

At 7.75 cm^3, the concentration of the sodium salt of H = concentration of H.

(v) Deduce the K_a value for **H**, giving your reasoning and the units. [2]

At max buffer capacity, pH = 4 (reading off the graph)

pH = pK_a
K_a = 10^{-4} = 1.00 x 10^{-4}

[Total: 20]